

THIRD SEMESTER--B.TECH- Course Work
END SEMESTER EXAMINATION, NOV- DEC, 2022

Course Code-ECMTC04

Course Title- Mathematics for Signal Processing and Communication

Time- 3:00 hours

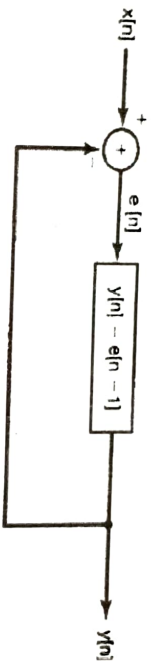
Max. Marks- 50

Note:- Attempt all the five questions. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Attempt any 2 parts of the questions as below	Marks	CO
1a	Define order of an element of a group G . Prove that a group G is abelian if and only if $(ab)^{-1} = a^{-1}b^{-1}, \forall a, b \in G$	5	CO1
1b	Find the order of each element in the group $G = \{a, a^2, a^3, a^4, a^5, a^6 = e\}$, where e being identity element of the group. Also give the example of a finite abelian group which is not cyclic. Justify your answers.	5	CO1
1c	Show that the set $G = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$ with binary operation addition and multiplication is a field.	5	CO1
2a	The intersection of any two subspace W_1 and W_2 of a vector space $V(F)$ is also a subspace of $V(F)$. Prove or disprove union of two subgroup is subgroup.	5	CO2
2b	If v_1, v_2, v_3 are linearly independent vectors of $V(F)$, where F is the field of real numbers, then show that $v_1 + v_2, v_1 - v_2, v_1 - 2v_2 + v_3$ is also linearly independent.	5	CO2
2c	Let $T: \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$ be a linear transformation defined by $T(x, y, z) = (x - y + 2z, 2x + y, -x - 2y + 2z)$, Find the basis and dimension of Range space of T and Null space of T . Also Verify Rank- Nullity theorem.	5	CO2
3a	(i) Prove that $\nabla^2 \left(\frac{x}{r^3} \right) = 0$. (ii) Prove that $\nabla^2 f(r) = f''(r) + \frac{2}{r} f'(r)$.	5	CO3
3b	Verify Green's theorem in the xy -plane for $\oint_C [(2xy - x^2)dx + (x^2 + y^2)dy],$ where C is the boundary of the region enclosed by $y = x^2$ and $y^2 = x$.	5	CO3
3c	Evaluate the surface integral $\iint_S \mathbf{F} \cdot \mathbf{n} dA$ where $\mathbf{F} = z^2 \mathbf{i} + xy \mathbf{j} - y^2 \mathbf{k}$ and S is the portion of the surface of the cylinder $x^2 + y^2 = 36, 0 \leq z \leq 4$ included in the first octant.	5	CO3

4a	If $f(z) = u + iv$, is any analytic function of the complex variable z and $u - v = e^x(\cos y - \sin y)$, find $f(z)$ in terms of z	5	CO4
4b	Evaluate the following integral using Cauchy integral formula (i) $\int_C \frac{4-3z}{z(z-1)(z-2)} dz$, C is the circle $ z = \frac{3}{2}$. (ii) $\int \frac{e^z}{(z+1)^2} dz$, where C is the circle $ z-1 = 3$.	5	CO4
4c	Define removable and essential singularity with at least one example. Also find the residue of a function $f(z) = \frac{z^2}{(z+1)^2(z-2)}$ at its all poles.	5	CO4
5a	A tightly stretched string of length 2 meter with fixed end points is initially in a equilibrium position. It is set vibrating by giving each point a velocity $y_0 \sin^3(\frac{\pi x}{2})$. Find the displacement $y(x, t)$.	5	CO5
5b	An insulated rod of length 'b' has its ends A and B maintained at 0°C and 100°C respectively until steady state conditions prevail. If B is suddenly reduced to 0°C and maintained 0°C , find the temperature at a distance x from A at time t .	5	CO5
5c	Solve the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ subject to the conditions $u(0, y) = u(b, y) = u(x, 0) = 0$ and $u(x, a) = \sin(\frac{n\pi x}{2})$.	5	CO5

Attempt all questions. Missing data / information (if any) may be suitably assumed & mentioned in the answer.

Q. No.	Questions	Mar ks	CO
Q1	Attempt any 2 out of 3 parts		
a	The following are the impulse responses of discrete-time LTI systems. Determine whether each system is stable and/or causal. Justify your answers. (i) $h(n) = (-0.5)^n u(n) + (1.01)^n u(1-n)$ (ii) $h(n) = n(1/3)^n u(n-1)$	04	
b	Consider the feedback system as shown in Figure. Assume that $y[n] = 0$ for $n < 0$.  (i) Sketch the output $y[n]$ when $x[n] = \delta[n]$. (ii) Sketch the output $y[n]$ when $x[n] = u[n]$.	04	CO1
c	Determine whether or not each of the following continuous-time signals is periodic. If the signal is periodic, determine its fundamental period. i. $x(t) = [\cos(2t - \pi/3)]^2$ ii. $x(t) = \sum_{n=-\infty}^{n=+\infty} e^{-(2t-n)} u(2t-n)$	04	
Attempt any 2 out of 3 parts			
a	Consider the cascade of two LTI systems $h_1(n) = \sin(8n)$ and $h_2(n) = (a)^n u(n)$, $a \leq 1$, and where the input is $x(n) = \delta(n) - a\delta(n-1)$, Determine the output $y(n)$.	04	

(iii) Find the convolution of two sequences $x(n) = (0, 5)^{n-2} u(n-2)$ and $h(n) = u(n+2)$.

04

c Suppose that $x(t) = 1, 0 \leq t \leq 1$ (0, otherwise) and $h(t) = x(t/\beta)$. (β lies between 0 and 1, $\beta \neq 0$). Find $y(t) = x(t) * h(t)$, and also plot $y(t)$.

04

Q3

Attempt any 4 out of 6 parts

a Given that $x(n)$ has discrete-time Fourier transform $X(e^{j\omega})$, express the Fourier transform of the following signal $g(n)$ in terms of $X(e^{j\omega})$.
 $g(n) = x(1-n) + x(-1-n)$

04

b Let $x(n)$ be a real & odd periodic signal with period $N = 7$ and discrete-time Fourier series coefficients X_k . Given that $X_{15} = j, X_{16} = 2j, X_{17} = 3j$. Determine the values of X_0, X_{-1}, X_{-2} and X_{-3} .

C03

04

i. Apply the appropriate Fourier transform properties to find the Fourier transform of $g(t) = t e^{-|t|}$.
ii. Use the result of part (i), along with the duality property, to determine the Fourier transform of $y(t) = \frac{4t}{1+t^2}$.

04

The output $y(t)$ of a causal LTI system is related to the input $x(t)$ by the equation.

d $\frac{dy(t)}{dt} + 10y(t) = \int_{-\infty}^{\infty} x(\tau)z(t-\tau)d\tau - x(t)$ where $z(t) = e^{-t}u(t) + 3\delta(t)$.

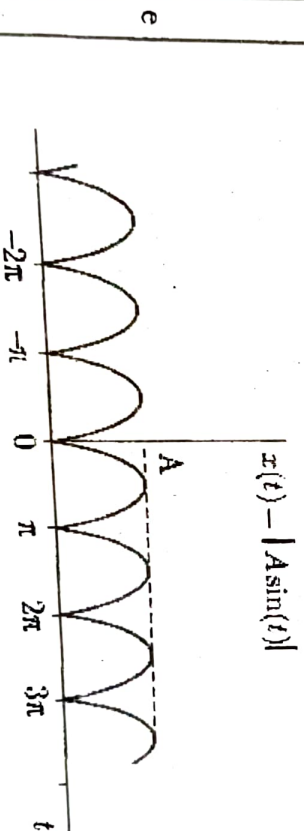
04

(i) Find the frequency response of the system.

(ii) Find the impulse response of the system.

Find the exponential Fourier series for the full-wave rectified sine wave depicted below.

C03



04

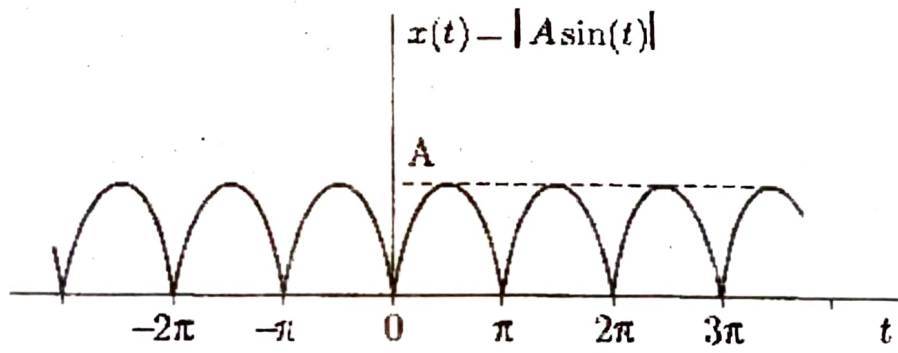
e Let $x(t)$ be a periodic signal with fundamental period T and Fourier series coefficients X_n . Derive the Fourier series representation of each of the following signals in terms of X_n .

04

(i) $\frac{d^2 x(t)}{dt^2}$ (ii) $x(2t - 2)$

Attempt any 2 out of 3 parts

Given the transfer function $H(s) = \frac{1}{(s+a)(s+b)}$ of an LTI system - If $h(t)$ were causal, over what range of values of a' and b' would the system be BIBO stable? State the ROC. - If $h(t)$ were two-sided, over what range of values of a' and b' would the system be BIBO stable? State the ROC. - If $h(t)$ were non-causal, over what range of values of a' and b' would the system be BIBO stable? State the ROC.	04	
A stable system has input $x(t)$ and output $y(t)$, where $x(t) = u(t)$ & $y(t) = e^{-t} \cos(2t)u(t)$. Use Laplace transforms to determine the transfer function and impulse response of the system.	04	CO4
Consider the signal $x(t)$ given below, - Express $x(t)$ in terms of basic functions. - Find out the Laplace transform of $x(t)$.	04	
		
Q5 Attempt any 2 out of 3 parts		
A system is described by the difference equation $y[n] - y[n-1] + \frac{1}{4}y[n-2] = x[n] + \frac{1}{4}x[n-1] - \frac{1}{8}x[n-2]$. Find the transfer function of the inverse system. Does a stable and causal inverse system exist? If yes, what is its impulse response?	05	
Given the following five facts about a discrete-time signal $x[n]$ with Z-transform $X(z)$: $x[n]$ is real and right-sided. $X(z)$ has exactly 2 poles. $X(z)$ has exactly 2 zeros at the origin. $X(z)$ has a pole at $z = \frac{1}{2}e^{j\pi/3}$. $X(1) = 8/3$. Determine $X(z)$ and specify its ROC.	05	CO5
Given the z-transform pair, use Z-Transform properties to determine the time-domain signals corresponding to the following z-transforms. Consider appropriate ROCs: $Y(z) = X(z^{-1})$ $Y(z) = \frac{d}{dz} X(z)$ $Y(z) = \frac{z+z^{-1}}{2} X(z)$	05	

b	(i) Show that the convolution of an odd and even function is an odd function. (ii) Find the convolution of two sequences $x(n) = (0.5)^{n-2}u(n-2)$ and $h(n) = u(n+2)$.	04	C03
c	Suppose that $x(t) = 1, 0 \leq t \leq 1$ (0, otherwise) and $h(t) = x(t/\beta)$, (β lies between 0 and 1, $\beta \neq 0$). Find $y(t) = x(t) * h(t)$, and also plot $y(t)$.	04	
Q3 Attempt any 4 out of 6 parts			
a	Given that $x(n)$ has discrete-time Fourier transform $X(e^{j\omega})$, express the Fourier transform of the following signal $g(n)$ in terms of $X(e^{j\omega})$. $g(n) = x(1-n) + x(-1-n)$	04	C03
b	Let $x(n)$ be a real & odd periodic signal with period $N = 7$ and discrete-time Fourier series coefficients X_k . Given that $X_{15} = j, X_{16} = 2j, X_{17} = 3j$. Determine the values of X_0, X_{-1}, X_{-2} and X_{-3} .	04	
c	i. Apply the appropriate Fourier transform properties to find the Fourier transform of $g(t) = t e^{- t }$. ii. Use the result of part (i), along with the duality property, to determine the Fourier transform of $y(t) = \frac{4t}{1+t^2}$.	04	
d	The output $y(t)$ of a causal LTI system is related to the input $x(t)$ by the equation. $\frac{dy(t)}{dt} + 10y(t) = \int_{-\infty}^{\infty} x(\tau)z(t-\tau)d\tau - x(t)$ where $z(t) = e^{-t}u(t) + 3\delta(t)$. (i) Find the frequency response of the system. (ii) Find the impulse response of the system.	04	C03
e	Find the exponential Fourier series for the full-wave rectified sine wave depicted below. 	04	
f	Let $x(t)$ be a periodic signal with fundamental period T and Fourier series coefficients X_n . Derive the Fourier series representation of each of the following signals in terms of X_n . (i) $\frac{d^2 x(t)}{dt^2}$ (ii) $x(2t-2)$	04	

c:

Roll No.

III SEMESTER B.Tech (IT, ITNS)

END-SEMESTER EXAMINATION 2022

Use Code-INECC06/ITECC06

Course Title-Probability & Stoch. Proc.

ie: 3:00 Hrs

Max Marks: 50

Attempt all questions. Missing data / information (If any) may be suitably assumed & mentioned in the answer.

Questions	Marks	CO
-----------	-------	----

Attempt any 2 out of 3 parts

a	Given that two events A_1 and A_2 are statically independent, show that i. A_1 is independent of $\bar{A}_2$ ii. $\bar{A}_1$ is independent of A_2 iii. $\bar{A}_1$ is independent of $\bar{A}_2$	05	CO1
b	A random experiment has sample space $S = \{a, b, c\}$. Suppose that $P[\{a, c\}] = 5/8$ and $P[\{b, c\}] = 7/8$. Use the axioms of probability to find the probabilities of the elementary events.	05	
c	i. Find $P[A/B]$ if $A \cap B = \emptyset$, if $A \subset B$, and if $B \subset A$ ii. Show that if $P[A/B] > P[A]$, then $P[B/A] > P[B]$. Comment with proper justification whether the statement is TRUE or FALSE.	05	

Attempt any 2 out of 3 parts

a	The CDF of a discrete random variables X is given as follows: $F_X(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{4} & 0 \leq x \leq 1 \\ \frac{1}{2} & 1 \leq x \leq 3 \\ \frac{5}{8} & 3 \leq x \leq 4 \\ 1 & x \geq 4 \end{cases}$	05	CO2
ii.	Determine the value of (i) $P[X < 2]$ and (ii) $P[0 \leq X \leq 4]$		

The number N of calls arriving at a switchboard during a period of one hour has the PMF $P_N(n) = \frac{10^n e^{-10}}{n!}, n = 0, 1, \dots$			
b	i. What is the probability that at least two calls arrive within one hour? ii. What is the probability that at most three calls arrive within one hour?	05	
c	Let X and Y be identically distributed independent uniform random variables on $[0, 2\pi]$. Define a random variable as $Z = \cos(X + \beta \sin Y)$. Find (a) $E[Z]$ (b) σ_Z^2. Here 'β' is a constant.	05	

Q3

Attempt any 2 out of 3 parts

a	Let $Y = a_1 X_1 + a_2 X_2 + \dots + a_n X_n$ where $X_1, X_2, \dots, X_n$ are mutually independent. Show that $\phi_Y(t) = \phi_{X_1}(a_1 t) \phi_{X_2}(a_2 t) \dots \phi_{X_n}(a_n t)$	05	C03
b	Let X be a random variable with probability mass function $P\{X = r\} = \binom{n}{r} p^r (1-p)^{n-r}, \quad r = 0, 1, 2, \dots, n, \quad 0 \leq p \leq 1$ Find the PMFs of the RVs i. $Y = aX + b,$ ii. $(b)Y = X^2,$ and iii. $(c)Y = \sqrt{X}.$	05	
c	Consider the joint pdf of X and Y $f_{XY}(x, y) = \frac{1}{3\pi} e^{-\frac{1}{2} \left[\left(\frac{x}{3}\right)^2 + \left(\frac{y}{2}\right)^2 \right]} u(x) u(y)$ Are X and Y independent random variables? Compute the probability of $\{0 < X \leq 3, 0 < Y \leq 2\}$.	05	C03

Q4

Attempt any 2 out of 3 parts

a	Let $Y(t) = X(t) - aX(t + d)$, where $X(t)$ is a WSS Gaussian random process. i. Find the mean and autocovariance of $Y(t)$. ii. Find the pdf of $Y(t)$. iii. Find the joint pdf of $Y(t)$ and $Y(t + s)$. iv. Show that $Y(t)$ is a Gaussian random process	05	CO4
---	--	----	-----

Let S_n denote a binomial counting process

- i. Show that $P[S_n = j, S_n = i] \neq P[S_n = j]P[S_n = i]$
- ii. Find $P\left[S_{n_2} = j / S_{n_1} = i\right]$, where $n_2 > n_1$
- iii. Show

$$\text{that } P\left[S_{n_2} = j / S_{n_1} = i, S_{n_0} = k\right] = P\left[S_{n_2} = j / S_{n_1} = i\right],$$

where $n_2 > n_1 > n_0$.

05

- c Write a detailed note on the Poisson Process. Derive the expression of the first and 'Nth' order joint pdf of the Poisson process. Is it a SSS process?

05

Attempt any 2 out of 3 parts

- a What do you understand by birth death process? Explain in detail.

05

- b i. For a homogeneous DTMC, Find the 'n' step probability transition matrix in terms of one step probability transition matrix.

05

- ii. Given a Markov chain $\{X_n, n \geq 0\}$, find the joint PMF $P(X_0 = i_0, X_1 = i_1, \dots, X_n = i_n)$.

CO5

- c i. A 2-state homogeneous DTMC X_n is described by the state transition matrix $P = \begin{bmatrix} 1-p & p \\ q & 1-q \end{bmatrix}$. Find the PMF of the DTMC at time '3'. If the initial PMF vector is $[p_0 \ p_1]$.

05

- ii. Given that the DTMC is in state '0' at time '5', what is the probability that the random process shall be in state '1' at time '8'.

NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY

THIRD SEMESTER-B.TECH(ECE & EIoT)

Course Code: ECECC06/ EIECC06
Course Title: Probability Theory and Stochastic Process

Time: 3:00 Hours

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Max. Marks: 50

Question		Marks	CO
Attempt any 2 out of 3 parts			
(a)	A lot of 100 semiconductor chips contains 20 that are defective. Two chips are selected at random, without replacement, from the lot. (a) What is the probability that the first one selected is defective? (b) What is the probability that the second one selected is defective given that the first one was defective? (c) What is the probability that both are defective?	5	CO1
(b)	Suppose a person is trapped in a funhouse at a place containing three doors. The first door leads to the outside after 5 minutes of travel. The second door leads to a twisty tunnel that comes back to the starting spot after 20 minutes. The third door is similarly twisty turny and leads back to the same spot after 30 minutes. If the person is equally to choose any one of the three doors whenever presented with the choice, what is the expected length of time until the person escape.	5	CO1
(c)	Suppose X has a moment generating function $m_X(t) = (1 - 2t)^{-\frac{1}{2}} \quad \text{for } t < \frac{1}{2}$ Find the first and second moment of X.	5	CO1
Attempt any 2 out of 3 parts			
(a)	Let X_1, X_2 be two independent random variables each of which has a probability density function of the form $f_{X_i}(x_i) = \frac{1}{(2\pi)^{1/2}} e^{-x_i^2/2}, \quad i = 1, 2 \quad -\infty < x_i < \infty$ Determine the mean and variance of Y, where $Y = \sum_{i=1}^2 X_i^2$	5	CO1
(b)	Let X and Y be Joint continuous random variables with joint PDF is given below $f_{XY}(x, y) = \begin{cases} 2 & x, y > 0 \text{ and } x + y \leq 1 \\ 0 & \text{otherwise} \end{cases}$ (a) Find co-variance of X and Y (b) Comment with proper justification whether X and Y are independent and uncorrelated.	5	CO1
(c)	Let Z be the random variable defined as $Z = \frac{\text{Max}(X,Y)}{\text{Min}(X,Y)}$. If X and Y are	5	CO1

	exponential density function with mean unity find the density function of Z.		
3	Attempt any 2 out of 3 parts (a) The lifetime of a cheap light bulb is an exponential random variable with mean 36 hours. Suppose that 16 light bulbs are tested, and their lifetimes measured. Use the central limit theorem to estimate the probability that the sum of the lifetimes is less than 600 hours. Express your answer either in terms of Q function or error function.	5	CO3
(b)	What is Chebyshev inequality. Prove it.	5	CO3
(c)	Let $Y = e^X$. If X is assumed to be a Gaussian random variable of zero mean and variance $1/8$. Find the density function of Y.	5	CO3
4	Attempt any 2 out of 3 parts (a) Let X(t) and Y(t) be both zero-mean and WSS random process. Consider the random process Z(t) defined by $Z(t) = X(t) + Y(t)$	5	CO4
(a)	(a) Determine the auto-correlation and the power spectrum of Z(t), if X(t) and Y(t) are joint WSS process (b) Show that if X(t) and Y(t) are orthogonal then the mean square of Z(t) is equal to the sum of the mean squares of X(t) and Y(t)	5	CO4
(b)	Suppose X(t) and Y(t) are zero mean wide sense stationary continuous time random process. If $Z(t) = cX(t)Y(t) + d$, where c and d are deterministic constant. Find the autocorrelation function of Z(t) in terms of autocorrelation functions of X(t) and Y(t). Assume X(t) and Y(t) are independent.	5	CO4
(c)	A random process is composed of sample functions of the form $X(t) = Ae^{-\alpha t}u(t - \Delta)$ where A is a Gaussian random variable with zero mean and variance σ^2, α is a constant and Δ is uniformly distributed between 0 and T and is independent of A. Comment with proper justification whether the random process X(t) is a wide sense stationary process or not.	5	CO4
5	Attempt any 2 out of 3 parts (a) The input X(t) to the RC filter shown in Figure below is a white noise process, then calculate the following	5	CO
(a)	(a) Power spectral density (PSD) of Y(t) (b) Auto-correlation $R_{XX}(\tau)$ of Y(t)		
(b)	Let X(t) be a zero mean wide sense stationary process with $R_X(\tau) = e^{- \tau }$. X(t) is the input to an LTI system with $ H(f) = \begin{cases} \sqrt{1 + 4\pi^2 f^2} & f < 2 \\ 0 & \text{otherwise} \end{cases}$ Find the following: $R_Y(\tau)$ $E[Y^2(t)]$	5	
c)	Enumerate the properties of autocorrelation function.	5	

Q.No.	Question	Marks	CO
Q 1	Attempt any two parts of the following		
1(a)	For the cascaded amplifier shown in Fig 1 Draw the small signal equivalent circuit and calculate overall voltage gain.	4	CO1
1(b)	Derive the voltage gain V_o/V_s at high frequency for a CE stage, as shown in Fig.2. Find a_1 and a_2 by time constant method.	4	CO1
1(c)	For the BJT differential amplifier shown in Fig.3, explain the large signal operation and derive expressions for emitter current I_E and calculate A_{DM} , A_{CM} and CMRR.	4	CO1
Q.2	Attempt any two parts of the following		
2(a)	Draw the block diagram of an op-amp and explain each block in detail with circuit diagram.	4	CO2
2(b)	Define the following electrical parameters of an op-amp:- 1) Slew rate 2) CMRR 3) Input offset voltage 4) Input bias current	4	CO2

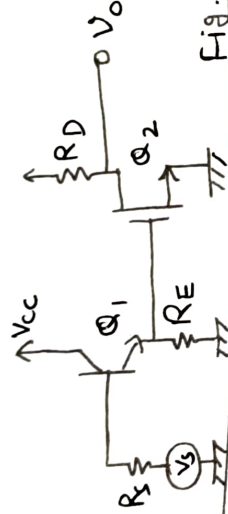


Fig. 1

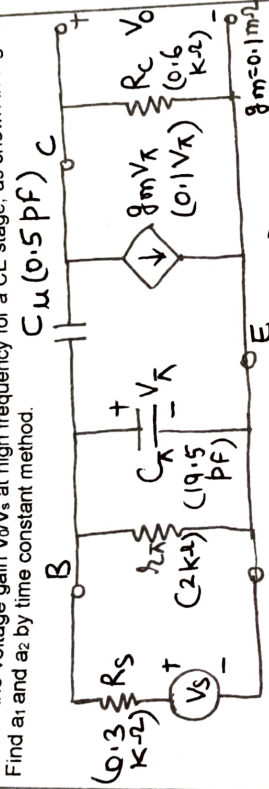


Fig. 2

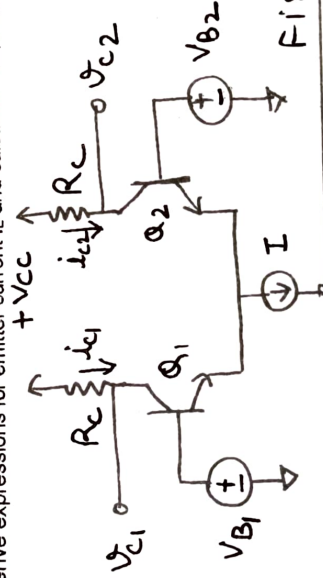
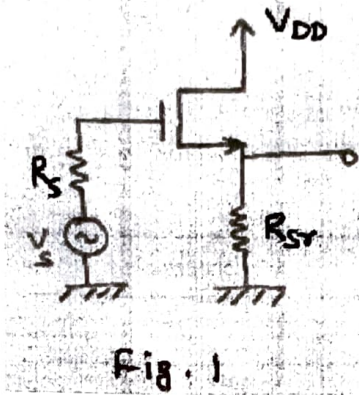
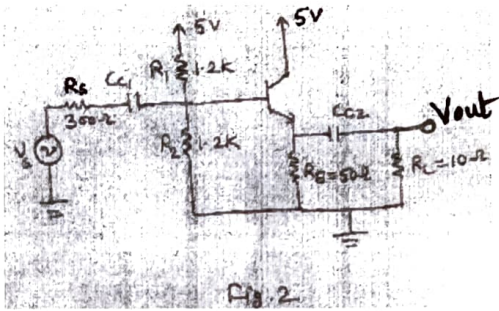
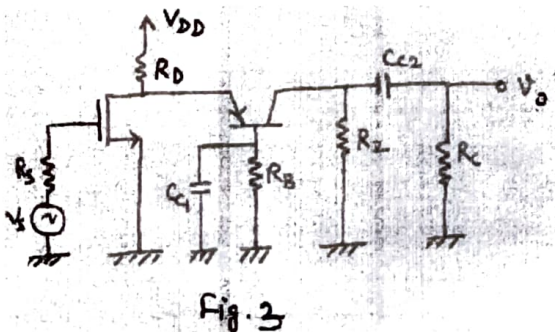
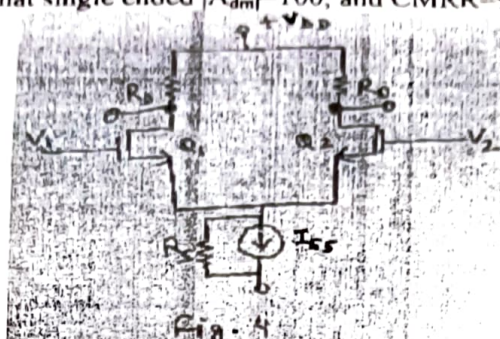
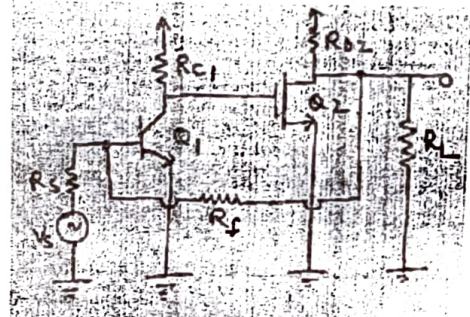


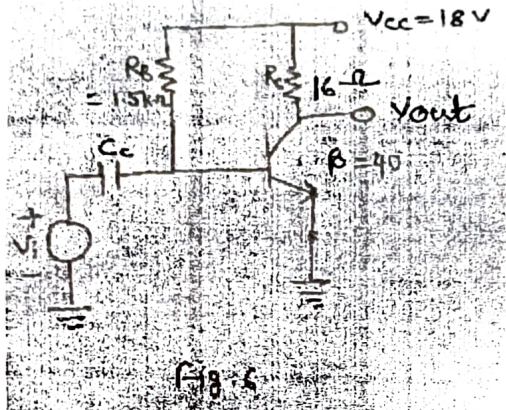
Fig. 3

31)	2(c)	Define operational transconductance amplifier and explain any one linear application of it	4	CO2
	Q.3	Attempt any two parts of the following		
	3(a)	Draw the circuit diagram of practical differentiator circuit and derive the relationship between input and output voltage. Also draw the frequency response for the same.	4	CO3
	3(b)	Design a second-order low-pass filter at a high cut-off frequency of 1 KHz and also draw the frequency response of the network.	4	CO3
	3(c)	Derive the expression for closed loop voltage gain for inverting and noninverting amplifier	4	CO3
	Q.4	Attempt any two parts of the following		
	4(a)	Derive the relationship between input and output voltage of antilog amplifier with circuit diagram.	4	CO4
	4(b)	Draw the circuit diagram of triangular wave generator and derive the expression of oscillation frequency for the same.	4	CO4
	4(c)	With the help of circuit diagram, describe the working of 555 timer as a monostable multivibrator. If the monostable multivibrator is to be used as a divide-by-2 network. The frequency of the input trigger signal is 2 KHz. If the value of $C=0.01\mu F$, what should be the value of R_A ?	4	CO4
	Q.5	Attempt any two parts of the following		
	5(a)	A class B output stage has to be designed to deliver an average power of 20 W to an $8\ \Omega$ load. The power supply V_{CC} should be 5 V greater than the peak output voltage. Determine, the supply voltage required, the peak current drawn from each supply, the total supply power and the power conversion efficiency. Also determine the maximum power that each transistor can dissipate safely.	4	CO5
	5(b)	Draw the circuit diagram of Wein bridge oscillator and also calculate the oscillation frequency.	4	CO5
	5(c)	Enumerate the effects of negative feedback on 1) gain 2) frequency response 3) distortion 4) noise.	4	CO5

Note: - Attempt all the five questions. All questions carry equal marks. Missing data/ information if any, should be suitably assumed & mentioned in the answer.

Q No.	Questions	Marks	CO
Q1	Attempt any two parts of the following:		
1(a)	Draw the circuit of cascode current mirror using BJT. Explain its working and find output current I_{out} and output impedance R_{out} .	4	CO1
1(b)	For the source follower circuit shown in Fig.1, find higher cut-off frequency f_H by time constant method. Assume that the dominant pole approximation is valid. Assume $r_o = \infty$.	4	CO2
 Fig. 1			
1(c)	Find the lower cut off frequency f_L for the following circuit (Fig.2) using short circuit time constant method.	4	CO2
 Fig. 2			
Q2	Attempt any two parts of the following:		
2(a)	For the cascaded amplifier shown in Fig.3. Draw small signal equivalent circuit and find input impedance R_{in} , output impedance R_o and voltage gain $G_v = V_o/V_s$.	4	CO2
 Fig. 3			

2(b)	With the help of large and small signal analysis of BJT differential amplifier, derive the expressions of collector currents i_{c1} and i_{c2} in terms of g_m , V_{in} and dc current I .	4	CO2
2(c)	The differential amplifier shown in Fig.4, has $r_d=50K$, $g_m=4\text{ mU}$ for each MOSFET. Determine R_d and R_s so that single ended $A_{dm} =100$, and $CMRR=90\text{dB}$.  Fig. 4	4	CO2
Q3	Attempt any two parts of the following:		
3(a)	Draw the circuit diagram of RC phase shift oscillator explain its working and derive the expressions for condition of oscillation the frequency of oscillation.	4	CO4
3(b)	For the circuit shown in figure (Fig.5) below, draw A circuit, B circuit and obtain the (a) input resistance R_{if}, (b) output resistance R_{of}, and (c) closed-loop voltage gain A_f.  Fig. 5	4	CO4
3(c)	Draw the two port representation of series -shunt feedback amplifier. Derive the expression for input impedance R_{if} and output impedance R_{of} series shunt feedback amplifier.	4	CO4
Q4	Attempt any two parts of the following:		
4(a)	Draw the circuit diagram of classical two stage CMOS operational amplifier with PMOS as primary transistors. Find Input common mode range (ICMR) for the first stage.	4	CO3
4(b)	Draw small signal equivalent circuit of classical two stage CMOS op-amp with PMOS as primary transistors and derive the expression for approximate voltage gain.	4	CO3
4(c)	Explain CMRR and PSRR and slew rate. Why compensation is required in the design of op-amp.	4	CO3
Q5	Attempt any two parts of the following:		
5(a)	Draw the circuit diagram of transformer coupled class A power amplifier. Draw its AC and DC load lines and find the maximum conversion efficiency for the same.	4	CO5
5(b)	Calculate the dc input power, ac output power and efficiency for the series fed amplifier circuit shown in Fig.6. Given that the input signal voltage results in a base current of 5 mA rms.	4	CO5



5(c)

Explain the harmonic distortion in case of large signal power amplifier. If the base current is given as $i_b = I_{bm} \cos \omega t$, and collector current is given as $i_c = G_1 i_b + G_2 i_b^2$, find B_0, B_1 and B_2 . Also explain the term total harmonic distortion (THD).

4

CO5

Max. Marks: 40

Attempt any TWO parts

Q No	Question	Mark s	CO
1	Attempt any TWO parts (a) A network has four inputs ABCD and four outputs WXYZ. ABCD represents a binary coded decimal digit. Further, WX represents the quotient and YZ represents the remainder when ABCD is divided by 4 (WX and YZ represent 2 bit binary numbers). Assume that the invalid inputs do not occur. Realize the network using NAND gates only.	4	CO1
	(b) Explain the following parameters: (i) Propagation Delay (ii) Noise Margin (iii) Fan-out (iv) Power dissipation	4	CO2
2	(c) Design and explain the NAND and NOR gate circuit implemented by CMOS transistors. What are the advantages and disadvantages of CMOS technology.	4	CO2
	Attempt any TWO parts (a) Using the Quine-McCluskey (Tabulation) method minimize the function given below having don't care terms $f(w, x, y, z) = \prod_{M}(0, 6, 7, 8, 9, 13) + \text{don't care}(5, 15)$	4	CO1
	(b) Design a code converter that converts 84-2-1 code into 8421 code and implement it by using four 4X1 MUXs and external gates.	4	CO3
	(c) Design an adder that adds two numbers of two bits each. Implement it by using complementary output decoder and AND/NAND gates.	4	CO3

3

Attempt any TWO parts

(a) Convert a positive edge triggered D flip-flop into a positive edge triggered AB flip-flop whose characteristic table is given as table 1.

A	B	$Q(t+1)$
0	0	0
0	1	$\overline{Q}(t)$
1	0	$Q(t)$
1	1	1

Table 1: Table for AB Flip-flop

(b) Using a negative edge triggered T flip flop design a synchronous mod-5 counter whose counting sequence is

$$100 \rightarrow 110 \rightarrow 111 \rightarrow 001 \rightarrow 011 \rightarrow 100$$

(c) Write short notes on PWM generator

4

CO3

4

CO3

4

CO3

4

Attempt any TWO parts

(a) Write VHDL code for 8X1 MUX.

(b) Write a VHDL code for a level triggered J-K flip-flop with clear and preset inputs.

(c) Explain in brief the various modeling styles in VHDL.

4

CO4

4

CO4

4

CO4

5

Attempt any TWO parts

(a) Implement the following functions using a cost efficient PLA assuming both true and complementary form are available at the output. Draw PLA table.

$$F_1(A, B, C, D) = \sum_m(0, 1, 2, 3, 6, 7)$$

$$F_2(A, B, C, D) = \sum_m(0, 1, 2, 3, 8, 9)$$

$$F_3(A, B, C, D) = \sum_m(0, 1, 6, 7, 14, 15)$$

(b) Explain in brief the various elements of the CPU.

(c) Design an FSM, using VHDL, that has an input w and an output z. The machine is a sequence detector that produces $z = 1$ when the previous two values of w were 00 or 11; otherwise $z = 0$.

4

CO5

4

CO5

4

CO5